Module 2: Basic Chemistry - Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Use atomic structure to explain basic bonding patterns.
2. Distinguish covalent, ionic, and hydrogen bonding in biological examples.
3. Explain why water properties matter for cells and organisms.
4. Interpret pH as a chemical condition that affects biology.
5. Track atoms through a simple chemical reaction.

Topics

- Atoms and elements
- Bonding and molecular shape
- Water as a biological solvent
- pH and chemical balance
- Chemical reactions in cells

Key Terms to Know

- **Atom** - Smallest unit of an element that keeps that element identity.
- **Element** - Pure substance defined by proton number.
- **Covalent bond** - Bond formed when atoms share electrons.
- **Ionic bond** - Attraction between oppositely charged ions.
- **Hydrogen bond** - Weak attraction involving polar hydrogen and electronegative atoms.
- **Polarity** - Uneven charge distribution across a molecule.
- **pH** - Measure of hydrogen ion concentration.
- **Buffer** - Chemical system that resists pH change.

Core Contents

1. Atoms form elements, and their electrons shape bonding behavior.
2. Covalent, ionic, and hydrogen bonds create different molecular properties.
3. Water polarity explains cohesion, temperature buffering, and dissolving ions or polar molecules.
4. pH tracks hydrogen ion concentration and affects biological molecules.
5. Cells depend on chemical reactions that rearrange matter without destroying atoms.

Study Tips

1. Start with the module's central claim before memorizing vocabulary.
2. Use the same-numbered lab as the evidence check for the module.
3. Explain one mechanism in words, then redraw it as a simple diagram.
4. Answer retrieval questions without notes, then revise with the terms list.

Connected Lab

- `lab-02_probability-statistics.md`

Generated Visuals

- Module 02: Basic Chemistry Evidence Map
- Module 02: Basic Chemistry Reasoning Sequence
- Module 02: Basic Chemistry Retrieval and Lab Check